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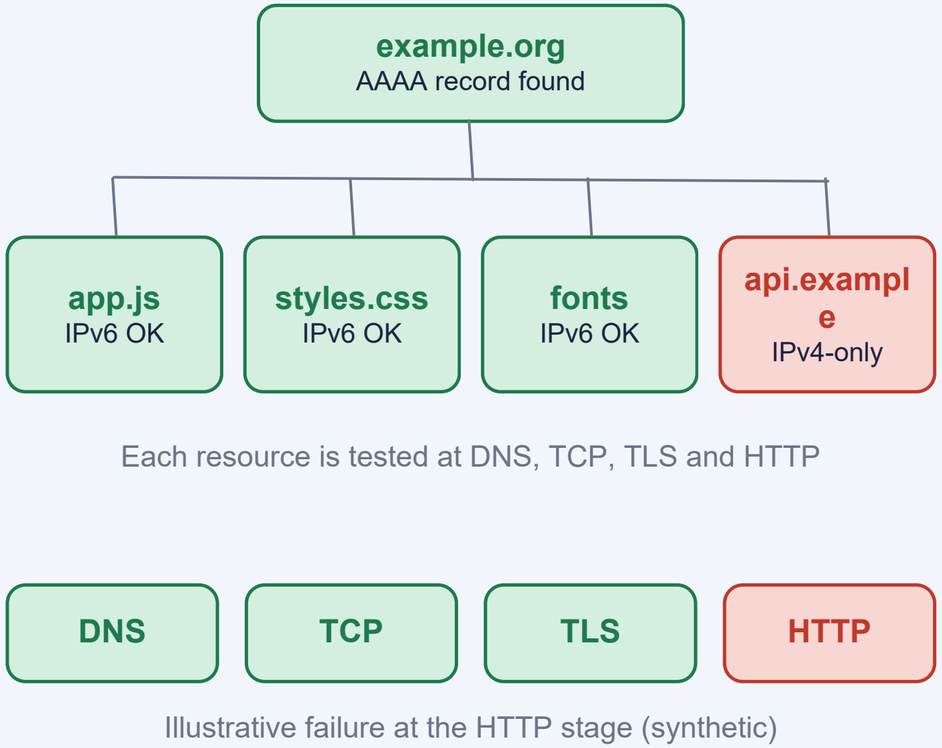
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THEME: Internet Measurement

IPv6 WEBSITE AND DEPENDENCY READINESS AUDITOR

Looking beyond the AAAA record.

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Each resource is tested at DNS, TCP, TLS and HTTP

Illustrative failure at the HTTP stage (synthetic)

PROBLEM DESCRIPTION

The Challenge of IPv6 Readiness Testing

A website is not fully IPv6-ready just because its apex domain resolves to an AAAA record. Modern pages depend on many external resources.

1. **Misleading AAAA Check**
   An AAAA record on the apex domain says nothing about the scripts, fonts, and APIs the page loads.

2. **Hidden IPv4-Only Dependencies**
   Third-party resources may remain IPv4-only and quietly break the page for IPv6-only users.

3. **Unclear Failure Stage**
   Failures can occur at DNS, TCP, TLS, or HTTP, and a simple pass/fail check cannot tell which.

4. **Manual, Non-Repeatable Testing**
   Mapping full dependency trees by hand is slow and hard to repeat across many sites.

The core problem: Teams need a reliable way to measure true IPv6 readiness across a page and its entire dependency tree.

BACKGROUND

Understanding IPv6 Reachability

Loading a modern web page triggers many requests to many hosts. Each one must work over IPv6 for the page to work on an IPv6-only network.

Apex Domain

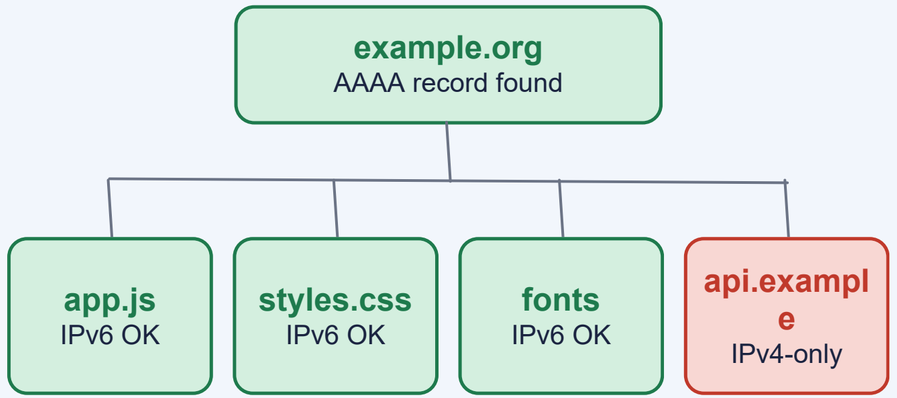
The main hostname of a site. An AAAA record shows IPv6 in DNS, not full readiness.

Page Dependencies

External scripts, stylesheets, fonts, images, and API endpoints, often hosted by third parties.

Failure Stages

Each resource passes through DNS resolution, TCP handshake, TLS negotiation, and HTTP retrieval. Any stage can fail.



Each resource is tested at DNS, TCP, TLS and HTTP

Illustrative dependency tree (synthetic)

LITERATURE & RFC REVIEW

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Standards That Guide Our Solution

Our approach builds on established Internet standards for IPv6, DNS, client connection behavior, and web resource measurement.

01 **RFC 8200: IPv6 Specification**
Defines the IPv6 protocol, the basis for testing reachability over IPv6.

02 **RFC 3596: DNS AAAA Records**
Defines the AAAA record that maps a hostname to an IPv6 address.

03 **RFC 8305: Happy Eyeballs v2**
Clients may fall back to IPv4, which can hide IPv6 failures from users.

04 **IETF webres6 Initiative**
Motivates measuring IPv6 readiness per web resource, not only per domain.

Our focus: Testing every resource of a page over IPv6 and pinpointing the exact stage where it fails.

PROPOSED SOLUTION

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From a Single URL to a Full Readiness Audit

We propose a browser-based auditor that loads a page, maps its dependencies, and tests each resource over IPv6 at every connection stage.



Key features:

- **Dependency mapping:** Records every script, stylesheet, font, image, and API call.
- **Stage-level diagnosis:** Pinpoints failure at DNS, TCP, TLS, or HTTP.
- **IPv6-only testing:** Checks each resource over IPv6 without IPv4 fallback.
- **Readiness scoring:** Rates each page and each dependency.
- **Repeatable reporting:** Exports results for all tested sites.

SOLUTION OPTIMIZATION

Manual / AAAA-Only Check vs. Our Proposed Solution

Our solution aims to replace shallow checks with a repeatable, dependency-aware audit that shows where IPv6 breaks.

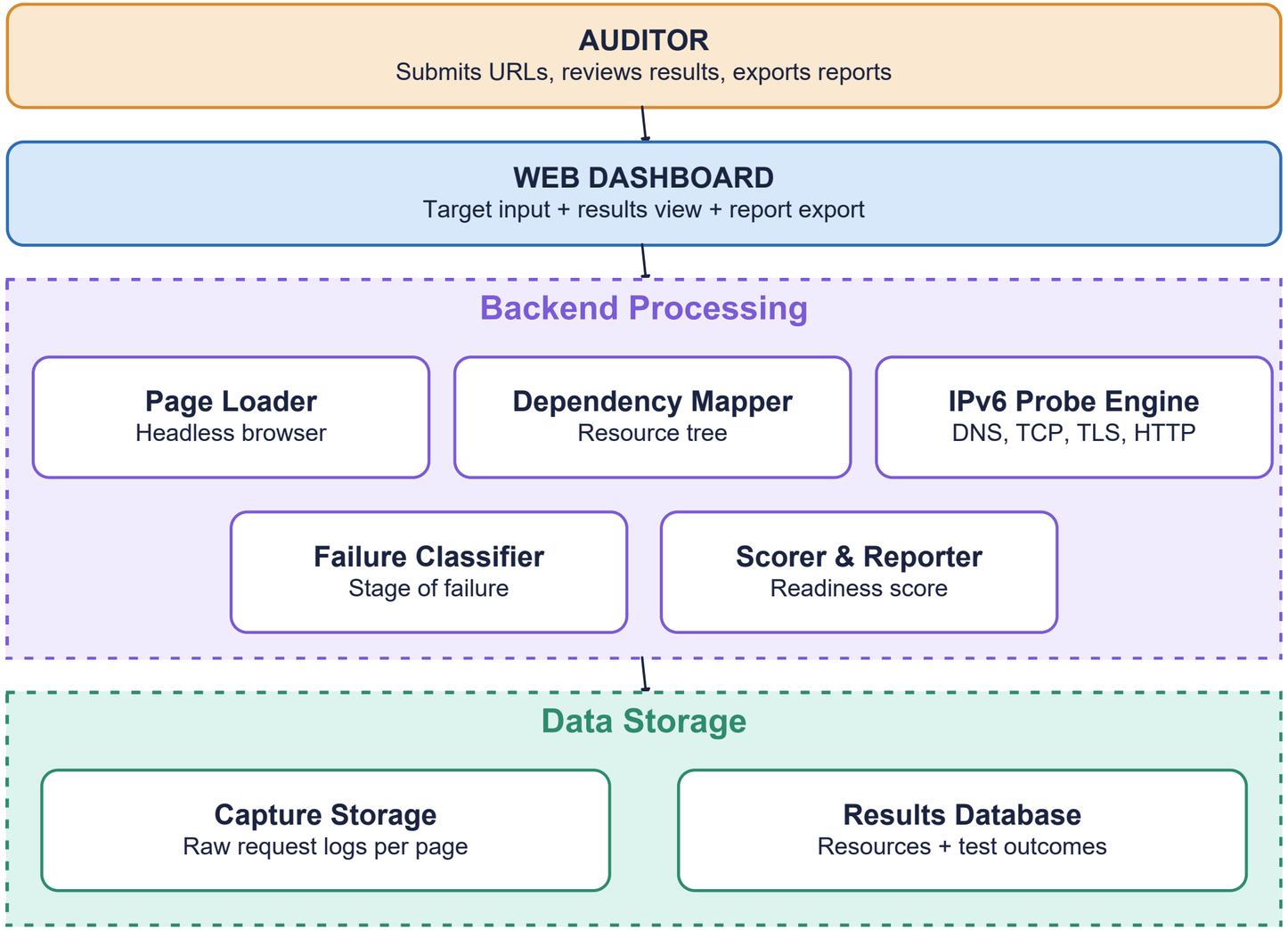
Parameter	Manual / AAAA-Only Check	Our Proposed Solution
Scope	Apex domain only	**Page plus full dependency tree**
Dependency Discovery	Manual inspection in browser tools	**Automatic resource mapping**
Failure Diagnosis	Pass or fail only	**Stage-level: DNS, TCP, TLS, HTTP**
Repeatability	Ad hoc, depends on the tester	**Same automated test on every site**
Scale	One site at a time	**Batch audit of 30 sites**
Reporting	Notes prepared by hand	**Structured readiness report**

ARCHITECTURE & DESIGN

How the System Works Internally

A web dashboard connects to a probe engine that tests every page resource over IPv6.

- **Web Dashboard:** Enter targets, view results.
- **Page Loader:** Captures the page's requests.
- **Dependency Mapper:** Builds the resource tree.
- **IPv6 Probe Engine:** Tests DNS, TCP, TLS, HTTP.
- **Failure Classifier:** Labels the failing stage.
- **Scorer & Reporter:** Rates readiness, exports.



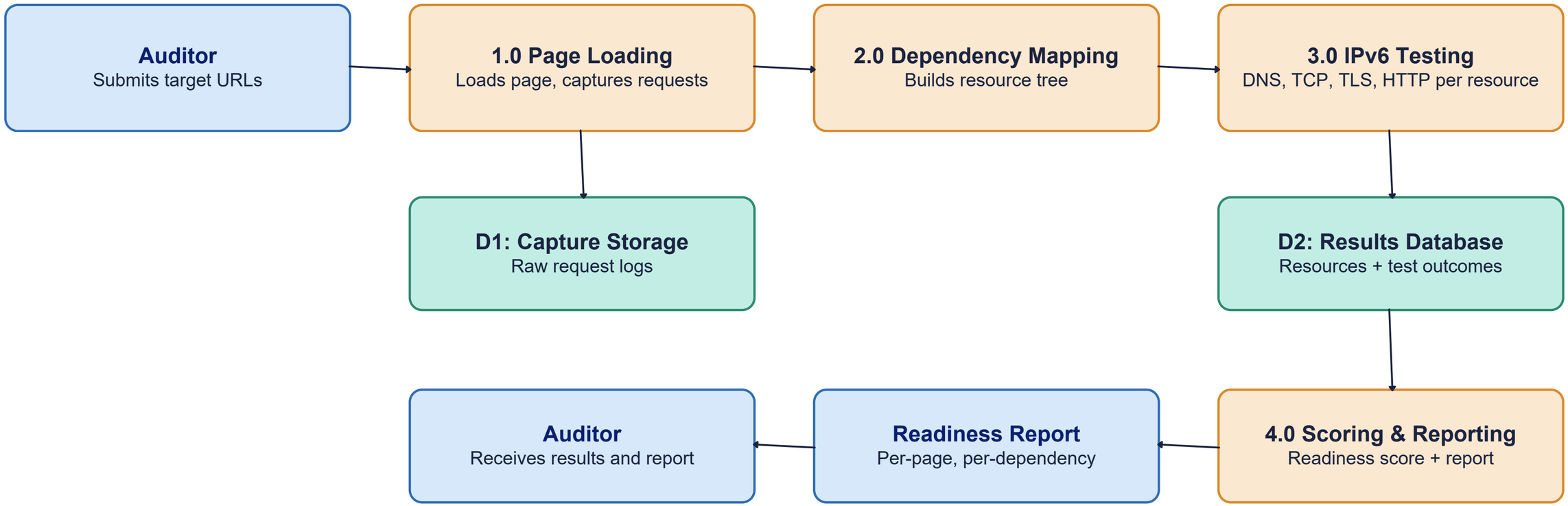
DATA FLOW DIAGRAM

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From Target URL to Readiness Report

The system turns a URL into a dependency tree, tests each resource over IPv6, and records the stage of any failure.



DEPLOYMENT TIMELINE

Our Implementation Roadmap

Our development plan follows four phases, from design to testing on real and controlled sites.

Phase 1: Foundation

Define requirements, select the 30 target sites, and design the architecture and IPv6 test environment.

Phase 2: Core Development

Implement page loading, request capture, dependency mapping, and IPv6 probes for DNS, TCP, TLS, and HTTP.

Phase 3: Reporting & Scoring

Build failure-stage classification, readiness scoring, dashboard views, and report export.

Phase 4: Testing & Deployment

Audit 30 sites and a controlled site seeded with IPv4-only assets, validate findings, and prepare deployment.

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THANK YOU

Looking beyond the AAAA record.

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Team ID: A3-PS027-TC126

Questions & Discussion